Homework 1
Physical Layer

Release Date 1403/12/26
Deadline 1404/1/20

Problem 1 (15 pts)

Compare the delay in sending an x-bit message over a k-hop path in a circuit-switched network and in a (lightly loaded) packet-switched network. The circuit setup time is s sec, the propagation delay is d sec per hop, the packet size is p bits, and the data rate is b bps. Under what conditions does the packet network have a lower delay? Also, explain the conditions under which a packet-switched network is preferable to a circuit switched network.

Problem 2 (10 pts)

There are two cellular networks to provide call service to subscribers, the first network has hexagonal cells and the second has square cells. There are 2100 separate frequency bands and each frequency band can transmit one T1 carrier. Assume that the use of similar frequency bands is prohibited in adjacent cells.

a) what is the maximum number of frequency bands that can be assigned to each cell in the first and second networks?

b) how many subscribers are served by each cell in the first and second networks?

Problem 3 (10 pts)

In Figure 1, the transmitted power is $125\ mW$, and the minimum acceptable SNR at the receiver is $30\ dB$

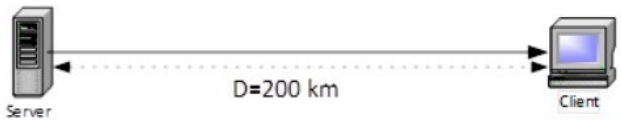


Figure 1: Server-Client setup.

a) Suppose the server is connected to the client with a Copper cable, and the transmission loss of the line is $2\ dB/km$. Is it possible to have a successful connection without using any amplifier?

b) Assume that we have m identical amplifiers, each with a gain of $10\ dB$. If your answer to the last part is no, find the minimum value of m to have a successful connection.

c) Repeat the previous parts assuming the server and the client are connected with an optical fiber, and the transmission loss of the line is $0.3\ dB/km$.

Problem 4 (15 pts)

Answer the questions below about Figure 2:

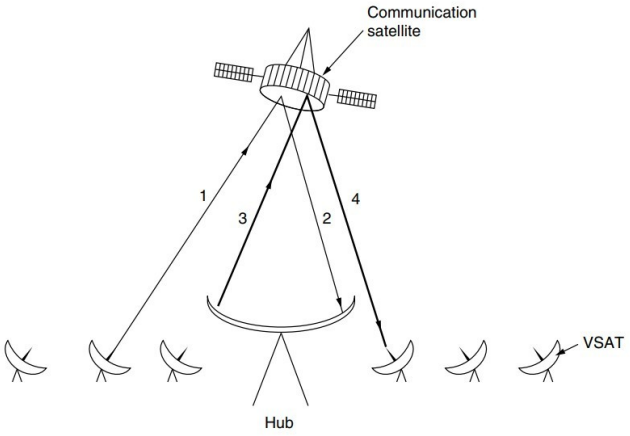


Figure 2: VSAT terminals and a hub.

a) How long will it take to transmit a 1-GB file from one VSAT to another using a hub? Assume that the uplink is $1\ Mbps$, the downlink is $7\ Mbps$, and circuit switching is used with $1.2\ sec$ circuit setup time.

b) Calculate the transmit time in the previous problem if packet switching is used instead. Assume that the packet size is $64\ KB$, the switching delay in the satellite and hub is $10\ \mu s$, and the packet header size is 32 bytes.

Problem 5 (10 pts)

Suppose that users A, B, C, and D are simultaneously transmitting bits using a CDMA system, with their assigned sequences as follows:

$$A = (-1, -1, -1, +1, +1, -1, +1, +1)$$
$$B = (-1, -1, +1, -1, +1, +1, +1, -1)$$
$$C = (-1, +1, -1, +1, +1, +1, -1, -1)$$
$$D = (-1, X, -1, -1, Y, Z, +1, -1)$$

a) Find X, Y and Z.

b) Is it possible that after all 4 users have sent their bits, the output becomes $(0, 0, -2, 0, -2, -4, 1, 2)$, if the answer is yes, determine the transmitted bit of each user.

c) Is it possible that after 3 users have sent their bits, the output becomes $(1, -3, 3, -1, 1, 1, 1, 1)$, if the answer is yes, determine the transmitted bit of each user.

Problem 6 (15 pts)

A simple telephone system consists of two end offices and a single toll office to which each end office is connected by a 1 MHz full-duplex trunk. The average telephone is used to make four calls per 8 hour workday. The mean call duration is 6 minutes. 10% of the calls are long distance (i.e., pass through the toll office).

a) What is the maximum number of telephones an end office can support? (Assume 4 kHz per circuit).

b) Explain why a telephone company may decide to support a lesser number of telephones than this maximum number at the end office.

Problem 7 (10 pts)

Consider a simple wireless network with three pair of transmitter and receivers, as shown in Figure 3. T_a, T_b, T_c are transmitters and their receivers are R_a, R_b, R_c, respectively. Suppose that A, B and C simultaneously send their data to their receivers. Also, consider that when a transmitter sends a signal with a power of P_t, the receiver gets a signal with a power of $P_r = \alpha \frac{P_t}{d^n}$, where d is the distance between the transmitter and the receiver in kilometers, α is a constant value and $n = 2$. If P_t for T_a, T_b and T_c are 1 ,2 and 3 watts, by considering that the noise is negligible, find the maximum achievable data rate at R_a, R_b and R_c. ($BW = 50MHz$)

- Note that in the presence of interference, to obtain the maximum data rate, SINR (Signal to Interference plus Noise Ratio) is used instead of SNR.

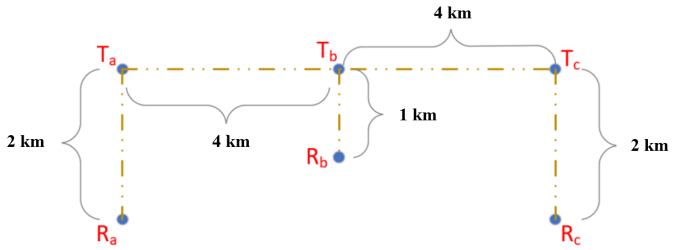


Figure 3: A network with 3 transmitters and 3 receivers.

Problem 8 (15 pts)

Figure 4 shows a router with a link that are part of a bigger network. Every time a packet is received at the router it is time-stamped (T_A). It is also time stamped when it is about to leave link L (T_B). When 5 packets of the size 200 bytes are transmitted through the link L, ($T_B - T_A$) is measured to be 6 ms, 5 ms, 2.4 ms, 4 ms, and 3.6 ms. When 5 packets of the size 500 bytes are transmitted through the link L, ($T_B - T_A$) is measured to be 11.0 ms, 6 ms, 4.2 ms, 3.8 ms, and 5.5 ms. Assume that processing delay at the router is negligible.

a) What is the average queuing delay experienced by the 200 and 500 byte packets?

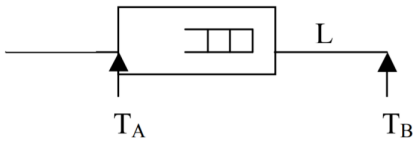


Figure 4: A router with link L.

b) What is the reasonable estimate of the total delay experienced by a packet of size 1000 bytes sent through the link L?

Submission

Please submit your answers to the problems in a `.pdf` file named **DN_HW1_studentID.pdf**, replacing `studentID` with your actual student ID.